

Course S&DAI

Nutri-MAS

Hybrid multi-agent system for personalized nutritional assistance

Acossi Giorgio Roberto - S5339095



**Università
di Genova**

DIBRIS DIPARTIMENTO
DI INFORMATICA, BIOINGEGNERIA,
ROBOTICA E INGEGNERIA DEI SISTEMI

Creative project

Proposed difficulty medium-hard



Project specification

Nutri-MAS is a conversational assistant that combines reactive agents, BDI agents, and LLM-based agents.

The system collects the user's profile, calculates indicative caloric targets, builds weekly plan, generates recipes, records the meals actually consumed, and rebalances the subsequent slots when the plan is modified.

Motivation

Nutritional assistance is a dynamic constraint-management problem, is not only recipe generation

User constraints

Calories
Diet
Allergens
preferences

Runtime Adaptation

Off-plan meals
Recalculation
Rebalancing

Planning constraints

Meal slots
Variety
Food frequencies

Reliable planning, flexible interaction, continuous adaptation

Technology stack



Qdrant



Streamlit



SPADE



Agent Speak



Python



Lite LLM



XMPP

Why a hybrid multi-agent system?

Each task is assigned to the decision-making model that best fits it

Reactive Agent

Reactive coordination

Gateway

BDI agents

Explicit reasoning and
control

Nutritionist · Planner · Chef

LLM agents

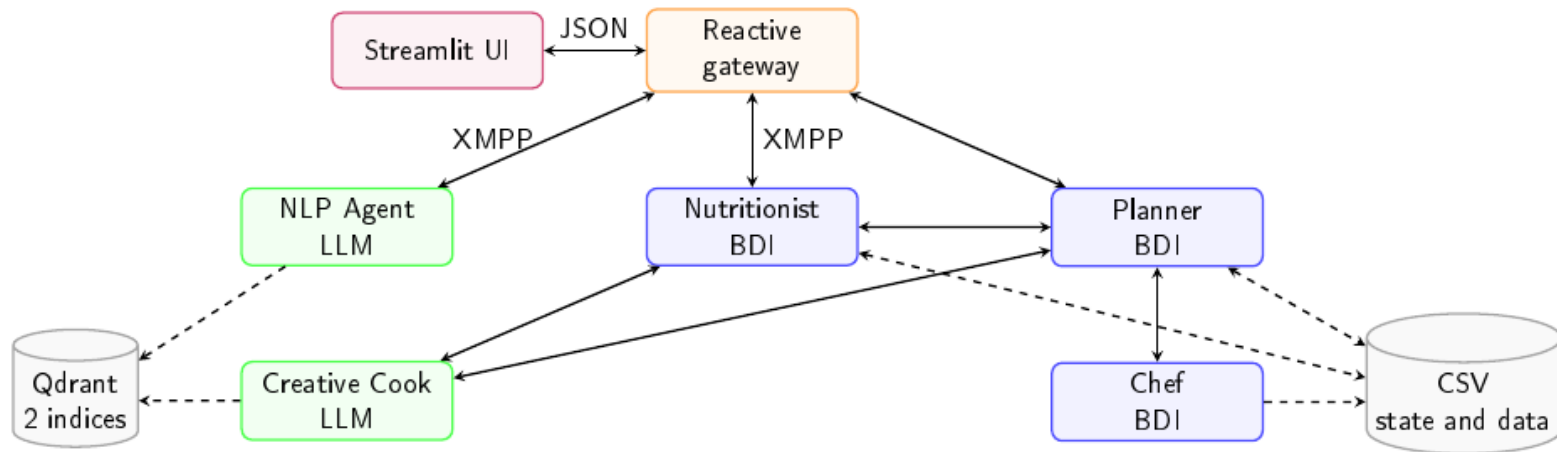
Flexible interpretation and
generation

NLP Agent · Creative Cook

Explicit rules provide control and LLMs provide flexibility

Nutri-MAS System Architecture

How reactive, BDI and LLM agents collaborate within Nutri-MAS



Agent	Responsibility
Gateway	Communication, validation and routing
NLP Agent	Natural language → AgentSpeak
Nutritionist	User state and daily monitoring
Planner	Weekly planning and local replanning
Chef	Valid template selection
Creative Cook	Recipe generation and meal estimation

Gateway

Reactive coordination between the user interface and the multi-agent system

Receive

- Receives user messages from the Streamlit UI
- Preserves the context of questions and answers
- Sends natural-language requests to the NLP Agent

Validate & Route

- Validates translated AgentSpeak commands
- Routes each command to the correct domain agent

Reply

- Receives structured responses from the agents
- Publishes results and system notifications to the UI

NLP Agent

Controlled translation from natural language to AgentSpeak commands

Interpret

- Receives natural-language requests from the Gateway
- Uses the previous question as context when interpreting short or ambiguous answers

Retrieve & select

- Searches Qdrant for semantically related linguistic templates
- Retrieves candidate message definitions from the registry
- Selects the message that best matches the user's intent

Build & validate

- Fills the arguments defined by the selected schema
- Validates syntax, arity, variables and destination agent
- Submits the resulting AgentSpeak command

NLP Agent

Natural language translation pipeline

```
username: <name>
message: <user text>
```

```
username: <name>
question: <question shown to the user>
answer: <user reply>
```

Input Context

Found template

```
{
  "nl_template": "I ate {actual_food} for {meal_type}.",
  "message": "nutritionist.log_meal"
}
```

```
{
  "id": "nutritionist.get_daily_recap",
  "agent": "nutritionist",
  "message_performative": "achieve",
  "description": "Show today's calorie recap.",
  "signature": "get_daily_recap(Uusername)",
  "schema": {
    "functor": "get_daily_recap",
    "arity": 1,
    "type": "object",
    "properties": {
      "Uusername": {
        "type": "string"
      }
    },
    "required": ["Uusername"]
  }
}
```

Performative schema

Nutrizionist

User state, nutritional targets and daily adaptation

Manage the user state

- Collects and updates the user profile
- Calculates nutritional goals, calories and macronutrient targets
- Maintains weight and meal histories

Monitor the daily plan

- Detects when a meal-slot time is crossed
- Requests confirmation of the planned meal
- Records confirmed meals or off-plan consumption

Adapt the plan

- Regenerates the weekly plan after relevant profile changes
- Triggers local rebalancing after an off-plan meal
- Coordinates Planner and Creative Cook when updates are required

Planner

Global weekly planning, recipe orchestration and local replanning

Build the weekly structure

- Converts nutritional targets into the meal slots
- Requests valid template domains from the Chef
- Choose templates while enforcing daily variety, weekly category frequencies and nutritional constraints

Materialize and activate the plan

- Sends selected templates to the Creative Cook
- Collects the generated recipes in a temporary draft

Replan individual meals

- Receives updated targets from the Nutritionist
- Rebuilds only future meals affected by a deviation
- Coordinates Chef and Creative Cook for the replacement

Chef

Deterministic filtering of valid meal templates

Maintain the template knowledge base

- Stores available meal templates as BDI beliefs
- Associates each template with diet type, meal slot, food category, calories and macronutrients

Filter valid candidates

- Applies explicit rules for diet, slot and nutritional targets
- Excludes templates already used or explicitly forbidden
- Supports guided requests for a specific food category
- Introduces variability only among valid candidates

Return planning candidates

- Provides the complete valid domain during weekly planning
- Selects one strict candidate during local replanning
- Returns an explicit failure when no valid template exists

Creative Cook

LLM-based recipe generation and off-plan meal estimation

Recipe generation

- Receives a validated meal template and nutritional targets
- Retrieves diet, allergens and culinary preferences
- Searches Qdrant for suitable ingredients
- Defines ingredients, quantities and preparation steps
- Estimates calories and macronutrients
- Returns a structured recipe to the Planner

Off-plan meal evaluation

- Interprets a freely described meal
- Searches for the complete dish or decomposes it into ingredients
- Estimates portions, calories and macronutrients
- Maps the meal to the closest available template
- Returns the nutritional estimate to the Nutritionist

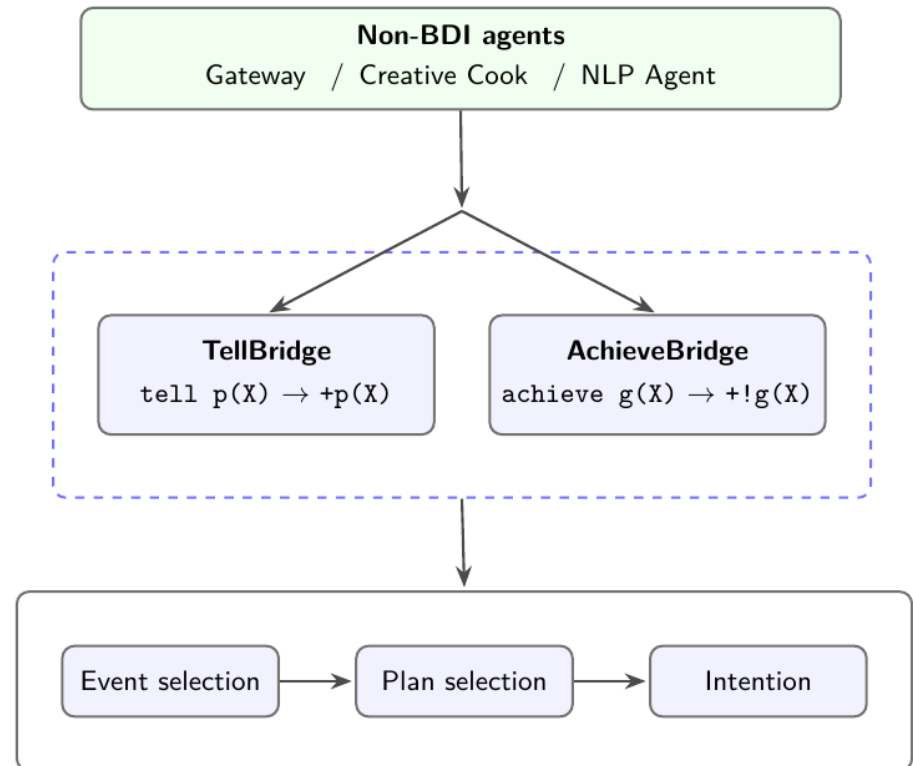
How do the agents communicate?

All agents exchange messages through **XMPP** while message bodies use **AgentSpeak-compatible syntax**.

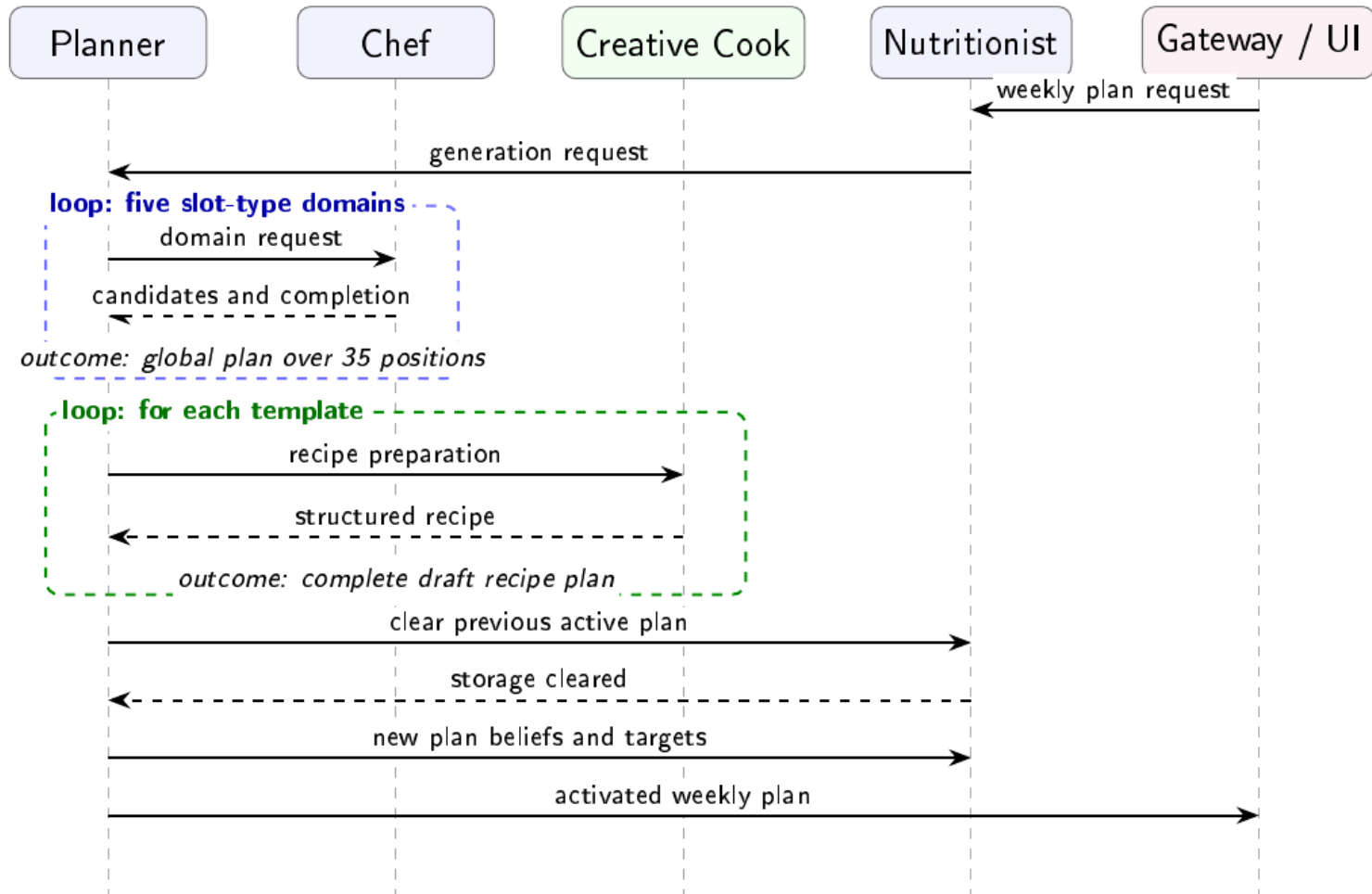
Adapting messages to the BDI model:

- **TellBridge**: converts a tell message into a belief update
- **AchieveBridge**: converts an achieve message into an achievement goal
- The resulting belief or goal becomes an event processed by the BDI reasoning cycle

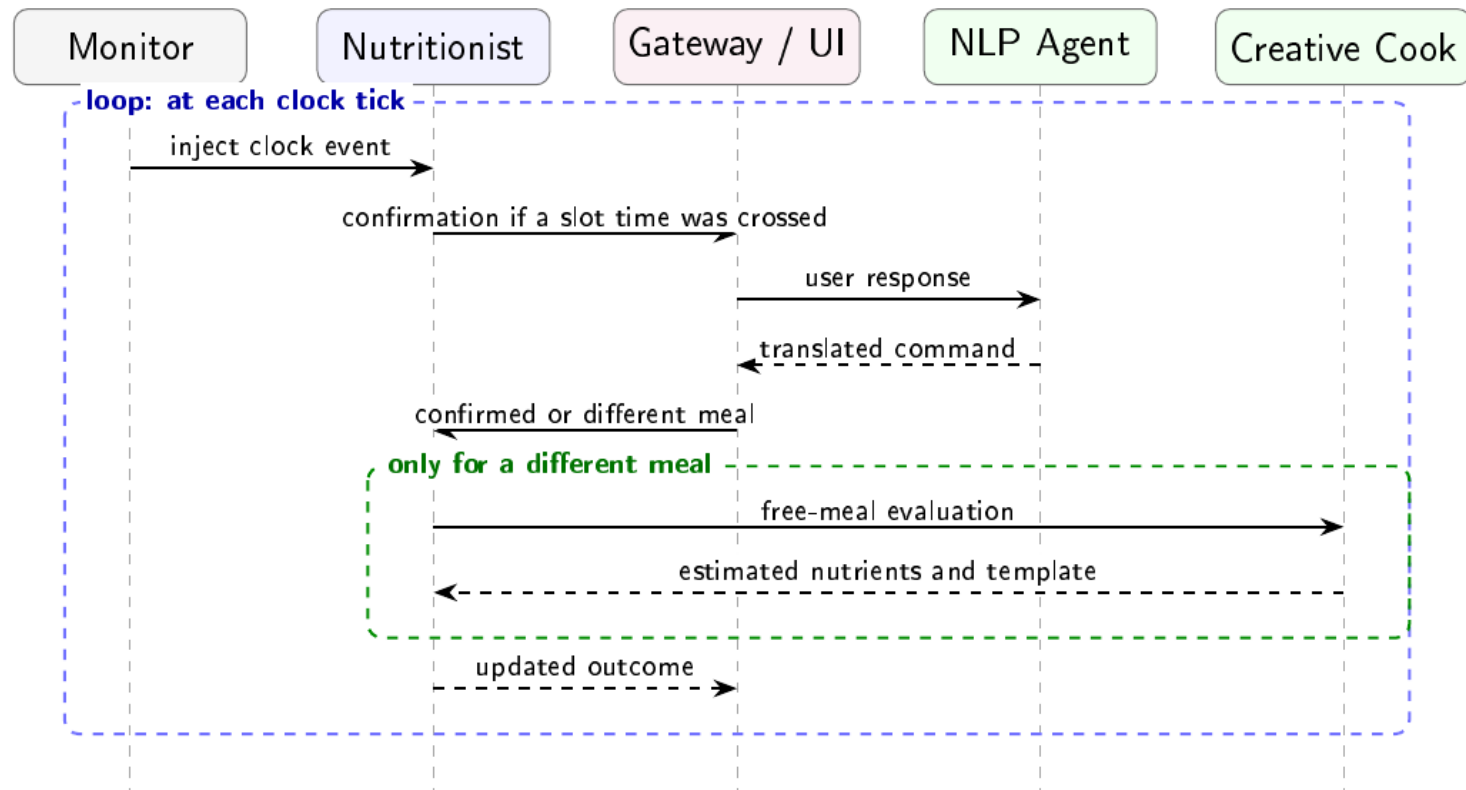
BDI agents communicate with non-BDI components through custom Python actions, which send standard XMPP messages.



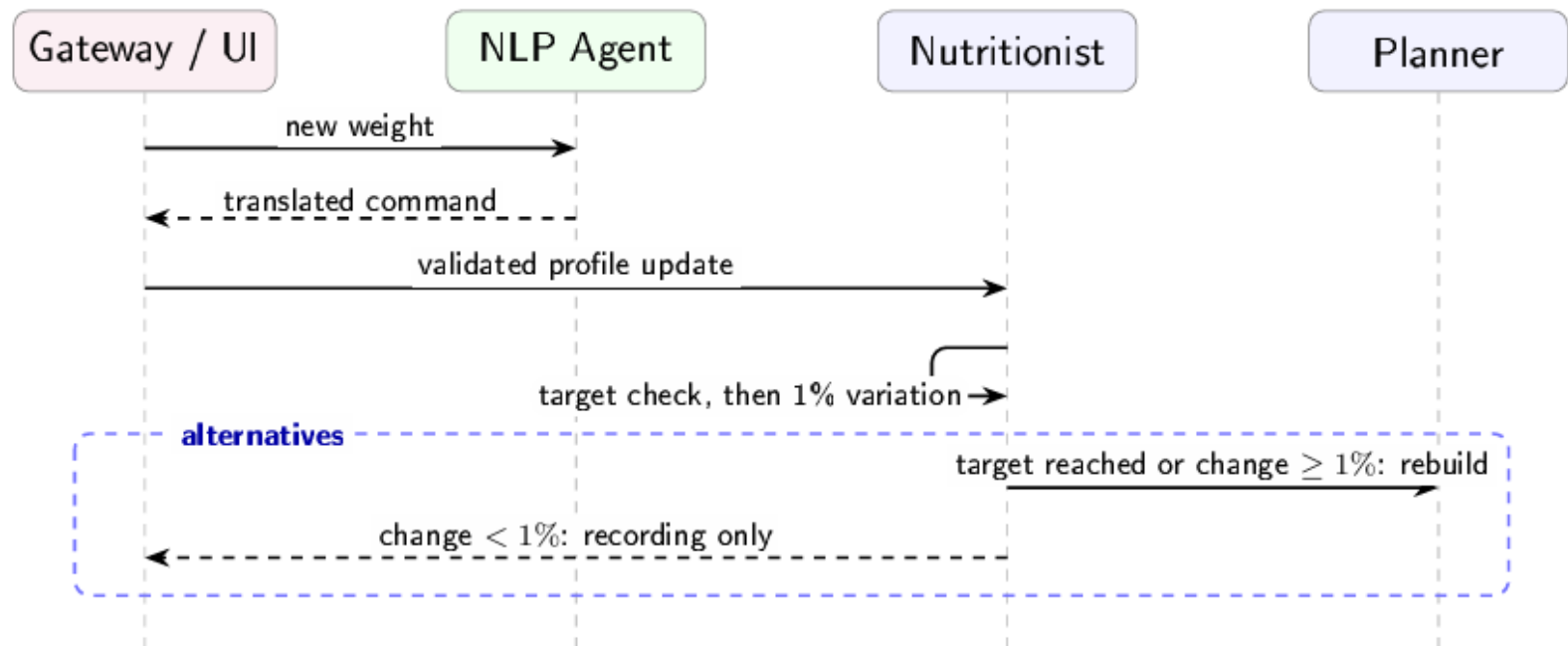
Weekly plan generation



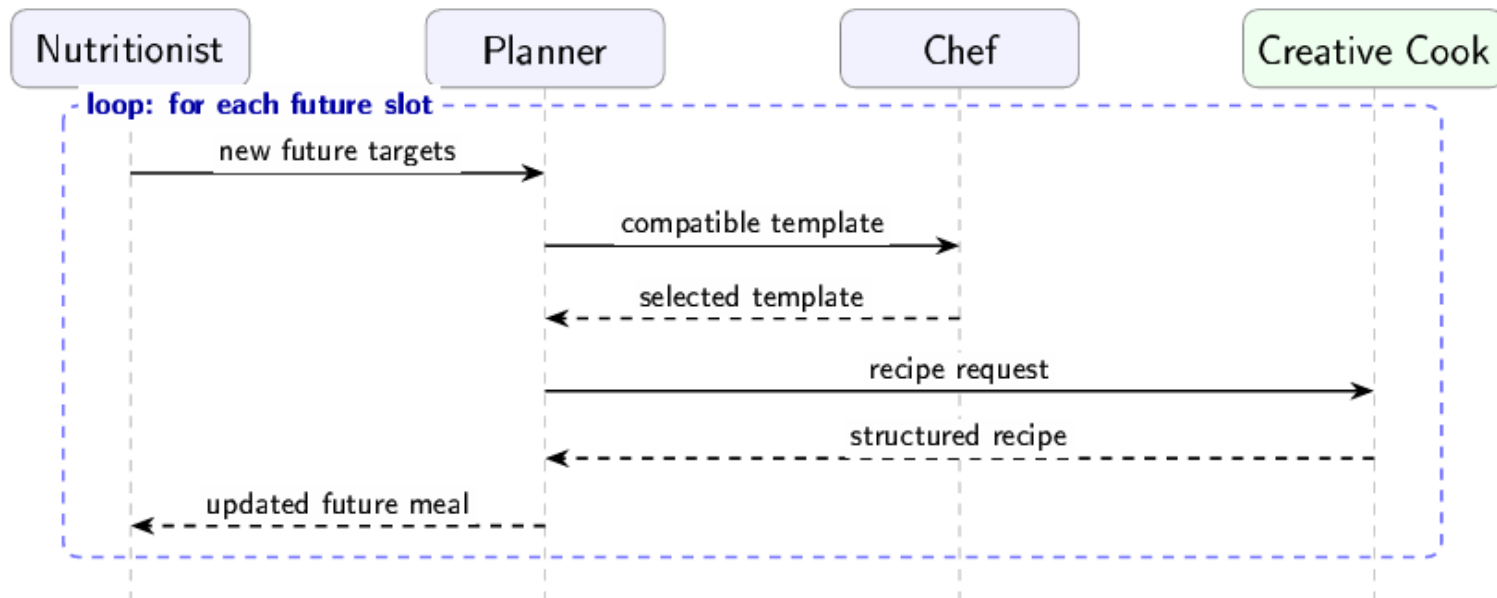
Meal Monitoring



Weight Update



Daily Plan Rebalancing



Data management

CSV

Persistent state

user, weight logs, meal logs

Materialized plans

weekly recipes and selected templates

Static knowledge

templates, nutrition rules, ingredients

Qdrant

Ingredients

semantic search over food names and
nutrients

Translations

user question mapped to performative

Demo

Onboarding

Rutni-MAS
What is your height in centimetres?
18:45

Rutni-MAS
What is your current weight in kilograms?
18:45

Rutni-MAS
How old are you?
18:47

Rutni-MAS
What is your gender? Choose male or female.
18:47

Rutni-MAS
What is your activity level? Choose sedentary, light, moderate, active, or very active.
18:47

You
173
18:45

You
93 kg
18:47

You
I'm 27
18:47

You
Male
18:47

You
moderate
18:48

Demo

Onboarding

Nutri-MAS
Do you have any allergies?
12:48

You
None
12:48

Nutri-MAS
Which diet do you follow? Choose omnivore, vegetarian, or vegan.
12:48

You
Omnivore
12:48

Nutri-MAS
Do you have culinary preferences or special recipe requests?
12:48

You
I prefer traditional Italian dishes
12:49

Nutri-MAS
Profile completed. Your daily target is 2359 kcal. Your goal is lose. The reference weight is 74.5 kg.
12:49

Nutri-MAS
Great, I am now preparing your weekly meal plan and recipes.
12:49

Demo

Week plan

You

What I need to eat this week

17:33

Nutri-MAS

Your current meal plan

17:34

> Monday

> Tuesday

> Wednesday

> Thursday

> Friday

> Saturday

> Sunday

Write a message...

↑

Demo

Daily plan

You

What i need to eat today17:27

Nutri-MAS

Your daily meal plan17:27

▼ Wednesday

Breakfast

Oat Yogurt Bowl with Banana and Nuts

603 kcal · Protein 34 g · Carbs 72 g · Fat 20 g

Ingredients

Oats: 85 g, High-protein yogurt: 300 g, Banana: 180 g, Almonds: 15 g, Walnuts: 10 g, Honey: 15 g

Description and preparation

Stir the oats into the yogurt and let them soften for a few minutes. Slice the banana on top, add the nuts, and finish with honey.

Morning snack

Crispbread with Cottage Cheese and Apple

242 kcal · Protein 15 g · Carbs 28 g · Fat 8 g

Ingredients

Wholemeal crispbread: 35 g, Cottage cheese: 70 g, Apple: 120 g, Pistachios: 8 g

Description and preparation

Spread the cottage cheese over the crispbread. Slice the apple and serve it on the side, then finish with the pistachios.

Demo

Week plan

You

What I need to eat this week

17:33

Nutri-MAS

Your current meal plan

17:34

> Monday

> Tuesday

> Wednesday

> Thursday

> Friday

> Saturday

> Sunday

Demo

Daily plan

Lunch

Chicken and Brown Rice Vegetable Bowl

651 kcal · Protein 44 g · Carbs 78 g · Fat 18 g

Ingredients

Chicken breast: 160 g, Brown rice: 90 g, Courgette: 200 g, Mixed leafy greens: 80 g, Carrot: 120 g, Olive oil: 14 g

Description and preparation

Cook the brown rice until tender. Grill the chicken and slice it. Lightly sauté the courgette, serve with the carrot and leafy greens, then top with olive oil and plate everything together.

Afternoon snack

Protein Yogurt with Banana, Oats and Almonds

242 kcal · Protein 19 g · Carbs 29 g · Fat 8 g

Ingredients

High-protein yogurt: 170 g, Banana: 100 g, Oats: 25 g, Almonds: 8 g

Description and preparation

Spoon the yogurt into a bowl. Top with sliced banana, oats and chopped almonds, then serve.

Dinner

Mozzarella with Boiled Potatoes and Salad

654 kcal · Protein 38 g · Carbs 79 g · Fat 22 g

Ingredients

Mozzarella: 150 g, Boiled potatoes: 350 g, Romaine lettuce: 100 g, Carrot: 100 g, Whole-wheat bread: 120 g, Olive oil: 5 g

Description and preparation

Warm the potatoes if needed and slice the mozzarella. Toss the lettuce and carrot with olive oil, then serve with the potatoes and bread.

Demo

Active Monitoring

The image shows a chat interface on a dark background. It features a series of messages from 'Nutri-MAS' and responses from 'You'. The messages are contained in rounded rectangular bubbles. The 'Nutri-MAS' messages are on the left, and the 'You' responses are on the right. Each message includes a timestamp in the bottom right corner.

Nutri-MAS
Did you eat the planned breakfast (Oat Yogurt Bowl with Banana and Nuts)?
17:01

You
Yes
17:16

Nutri-MAS
Breakfast is now marked as confirmed.
17:16

Nutri-MAS
Did you eat the planned lunch (Chicken and Brown Rice Vegetable Bowl)?
17:16

You
No, I ate a big pizza
17:17

Nutri-MAS
Big pizza was logged with 912 kcal. I'll rebalance the rest of your day now.
17:18

Nutri-MAS
Your daily rebalanced meal plan
17:18

Demo

Target Reached

You

My current weight is 74 kg

17:57

Nutri-MAS

You reached your target weight at 74 kg. I'm rebuilding your weekly plan for maintenance at 2364 kcal per day.

17:57

Nutri-MAS

Great, I am now preparing your weekly meal plan and recipes.

17:57

Nutri-MAS

Your weekly meal plan

18:08

> Monday

> Tuesday

> Wednesday

> Thursday

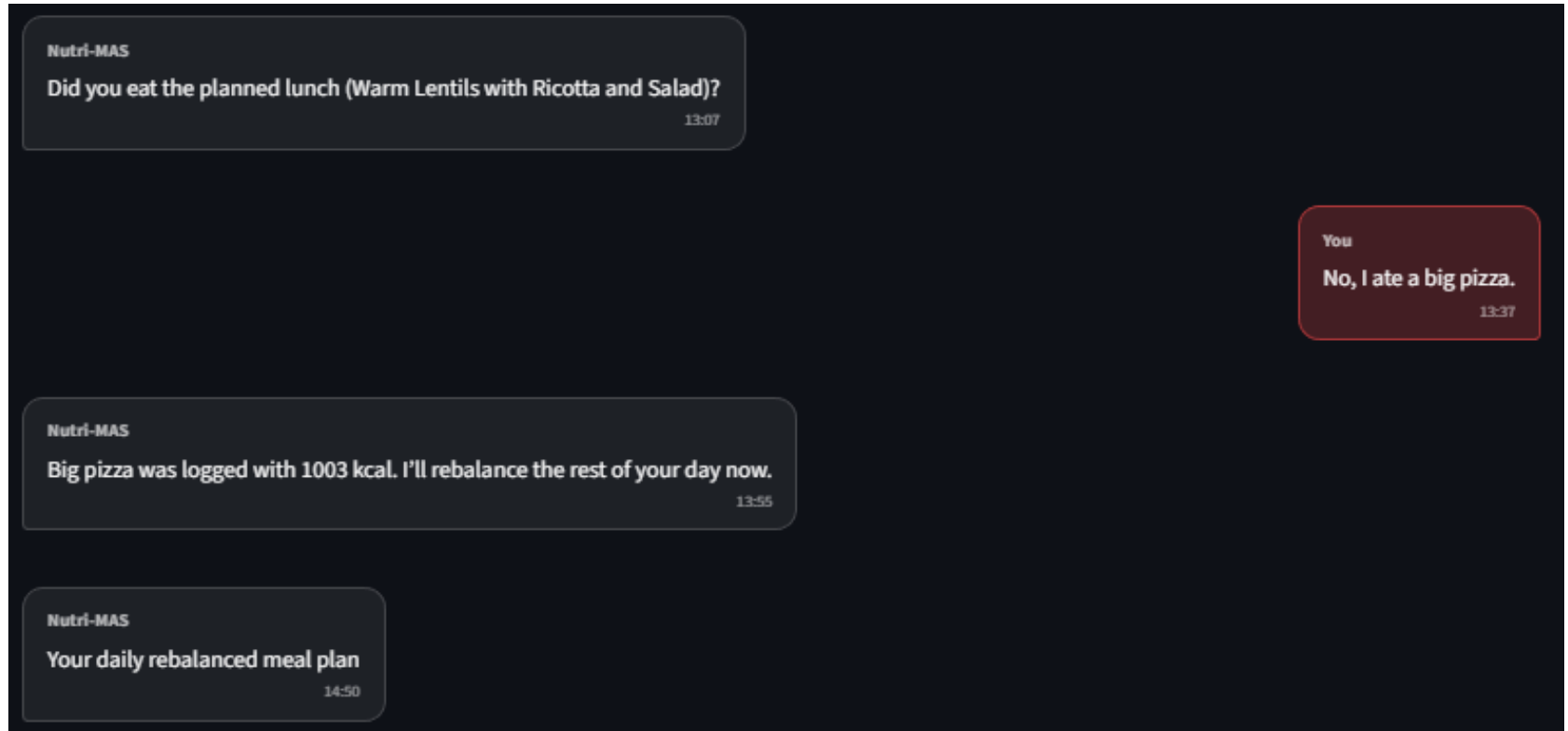
> Friday

> Saturday

> Sunday

Demo

Rebalancing



Demo

Rebalancing

Before

Lunch

Warm Lentils with Ricotta and Salad

506 kcal · Protein 31 g · Carbs 60 g · Fat 17 g

Ingredients

Boiled lentils: 250 g, Ricotta: 100 g, Whole-wheat bread: 100 g, Mixed salad greens: 100 g, Tomato: 150 g, Olive oil

Description and preparation

Warm the lentils and season them lightly. Serve with ricotta, sliced tomato, and salad greens dressed with olive oil.

Afternoon snack

Apple and Ricotta Snack

244 kcal · Protein 14 g · Carbs 28 g · Fat 17 g

Ingredients

Apple: 180 g, Ricotta: 100 g, Hazelnuts: 10 g

Description and preparation

Slice the apple and serve it with ricotta. Roughly chop the hazelnuts and sprinkle them on top.

Dinner

Whole-wheat pasta with ricotta and zucchini

507 kcal · Protein 30 g · Carbs 58 g · Fat 17 g

Ingredients

Whole-wheat pasta: 85 g, Ricotta: 170 g, Zucchini: 200 g, Olive oil: 10 g, Parmesan: 8 g

Description and preparation

Cook the pasta until tender. Sauté the zucchini with olive oil, then mix with ricotta and a little cooking water to moisten the pasta, finish with parmesan, and serve hot.

Rebalanced

Lunch

big pizza

1003 kcal · Protein 57 g · Carbs 88 g · Fat 40 g

Afternoon snack

Mini Vanilla Yogurt with Honey

63 kcal · Protein 3 g · Carbs 10 g · Fat 0 g

Ingredients

Apple: 180 g, Ricotta: 100 g, Hazelnuts: 10 g

Description and preparation

Slice the apple and serve it with ricotta. Roughly chop the hazelnuts and sprinkle them on top.

Dinner

Light Vegetable Velouté with Ricotta

170 kcal · Protein 10 g · Carbs 20 g · Fat 6 g

Ingredients

Whole-wheat pasta: 85 g, Ricotta: 170 g, Zucchini: 200 g, Olive oil: 10 g, Parmesan: 8 g

Description and preparation

Cook the pasta until tender. Sauté the zucchini with olive oil, then mix with ricotta and a little cooking water to moisten the pasta, finish with parmesan, and serve hot.

Limitations

LANGUAGE & KNOWLEDGE

- Finite linguistic templates
- Sometime the NLP Agent choose the wrong performative
- Manually encoded meal templates

NUTRITIONAL SAFETY

- Heuristic calorie and macronutrient targets
- Limited robustness of allergen and dietary constraint enforcement

PLAN OPTIMIZATION

- Rebalancing modifies only future meals of the current day
- Weekly constraints are not globally re-optimized

References

- **Mifflin et al. (1990)** Equation used to estimate the user's resting energy expenditure and daily caloric requirement.
- **Food Standards Australia New Zealand (2025)** Australian Food Composition Database used as the source of ingredient calories and macronutrients.
- **Centers for Disease Control and Prevention** Adult BMI categories used to determine the user's initial nutritional goal.

UniGe

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